

PARTICIPANT TRAINING NOTES

Digital Productivity

with Microsoft Power Platform & SharePoint

Power BI • Power Automate • Power Apps • SharePoint

LEARNING FOCUS

From manual processes to connected, low-code digital solutions

TWO-DAY HANDS-ON PROGRAMME

Participant manual • 2026 edition

PROGRAMME

Digital Productivity with Microsoft Power Platform & SharePoint

FORMAT

2 days • approximately 14 contact hours

TRAINER

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PARTICIPANT NAME _____

DEPARTMENT _____

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NOTE

Screens in Microsoft 365 are updated regularly. If a button label in your tenant differs slightly from this manual, look for the closest equivalent option or ask the trainer. Interface labels are written in **bold**, and formulas or expressions appear in `code format`.

ABOUT THIS MANUAL

Welcome to the programme

This manual accompanies a two-day, hands-on programme designed for managers, executives, non-executives and interns at PC MTBE. No prior experience with Microsoft Power Platform or SharePoint is required. You will start with the basic ideas, watch guided demonstrations, and then build a small working solution yourself.

Throughout the programme you will work on **one connected case study: the PC MTBE Process Improvement Request**. By the end of Day 2 you will have a SharePoint List that stores requests, a Power Apps Canvas App to submit and update them, Power Automate flows that notify and route approvals, and a Power BI report that shows request status and trends.

How to use this manual

- **Concept pages** explain each platform in plain language. Technical terms are explained the first time they appear and are collected in the Glossary (Appendix A).
- **Hands-on exercises** (terracotta banners) contain an objective, duration, prerequisites, numbered steps, the expected result, a checkpoint and common errors.
- **Callout boxes** highlight notes, trainer tips, troubleshooting advice and items to **check with your administrator**.
- **Notes areas** at the end of each section give you space to record your own observations and ideas.

Learning outcomes

By the end of this programme, you will be able to:

#	Learning outcome	Platform
1	Explain how SharePoint, Power Apps, Power Automate and Power BI work together to digitalise a business process.	All
2	Organise documents in a SharePoint site, use version history, and create a SharePoint List with suitable column types.	SharePoint
3	Build a Canvas App that views, submits and updates records in a SharePoint List, including simple validation.	Power Apps
4	Create automated flows for notifications, a basic approval and a scheduled reminder, and test them using run history.	Power Automate
5	Connect Power BI Desktop to a SharePoint List, prepare the data, and build a clear report on request volume, status, priority and trends.	Power BI
6	Describe how reports are published and packaged into a Power BI App, and identify when licensing or permissions apply.	Power BI
7	Identify at least one manual process in your own work that could be improved using these tools.	All

CHECK WITH YOUR ADMINISTRATOR

Before the programme, your IT team should confirm that each participant has: a Microsoft 365 account with access to SharePoint; access to Power Apps and Power Automate (make.powerapps.com and make.powerautomate.com); Power BI Desktop installed; and, if publishing will be practised, a Power BI Pro or Premium Per User licence or a workspace on Premium/Fabric capacity. A training SharePoint site where participants have **Edit** or **Owner** permission is strongly recommended.

TWO-DAY PROGRAMME AGENDA

Programme at a glance

Day 1 — Foundations, SharePoint and Power Apps

Time	Session	Activity type
8:30 – 9:00	Registration, welcome and programme objectives	Briefing
9:00 – 10:00	Section 1: How the four platforms work together; introduction to the case study	Presentation + discussion
10:00 – 10:15	Morning break	—
10:15 – 11:15	Section 2: SharePoint sites, document libraries, version history and permissions	Demonstration
11:15 – 12:30	Exercises 2.1 and 2.2: Build the request list and views	Hands-on
12:30 – 1:30	Lunch break	—
1:30 – 2:30	Section 3: Power Apps concepts, Power Apps Studio and Exercise 3.1	Demonstration + hands-on
2:30 – 3:30	Exercise 3.2: Customise, validate and test the app	Hands-on
3:30 – 3:45	Afternoon break	—
3:45 – 4:45	Exercise 3.2 continued; peer testing of apps	Hands-on
4:45 – 5:00	Day 1 recap and questions	Review

Day 2 — Power Automate, Power BI and integration

Time	Session	Activity type
9:00 – 9:15	Day 1 recap and warm-up quiz	Review
9:15 – 10:15	Section 4: Triggers, actions, conditions; Exercise 4.1	Demonstration + hands-on
10:15 – 10:30	Morning break	—
10:30 – 11:45	Exercises 4.2 and 4.3: Approval workflow and scheduled reminder	Hands-on
11:45 – 12:30	Section 5: Power BI concepts; Exercise 5.1 connect and prepare data	Demonstration + hands-on
12:30 – 1:30	Lunch break	—
1:30 – 3:00	Exercise 5.2: Build the Request Monitoring report	Hands-on
3:00 – 3:15	Afternoon break	—
3:15 – 3:45	Exercise 5.3: Publishing and the Power BI App (hands-on or demonstration)	Demonstration
3:45 – 4:30	Section 6: Integrated end-to-end exercise	Hands-on
4:30 – 5:00	Review questions, action planning and closing	Review

SECTION 1

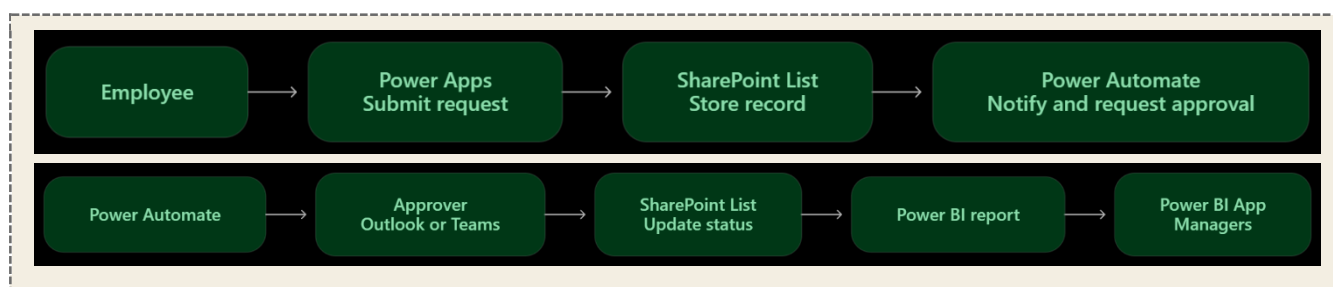
How the four platforms work together

Four tools, one connected solution: store, capture, automate and analyse.

Most organisations still rely on emails, spreadsheets and paper forms to manage everyday requests. Information is copied several times, follow-ups are manual, and managers wait for someone to compile a report. Microsoft 365 includes tools that connect these steps so that data is entered once and then reused automatically.

Microsoft Power Platform is a family of **low-code** tools — meaning you build solutions mainly by configuring screens and writing short formulas rather than traditional programming. In this programme, three Power Platform tools are combined with **SharePoint**, the Microsoft 365 service for storing and sharing information.

Platform	What it does	Role in our case study
SharePoint	Stores and shares documents and structured lists in team websites, with permissions and version history.	Holds the Process Improvement Requests list — the single source of data.
Power Apps	Builds business applications (forms and screens) that work in a browser and on mobile.	Provides a simple app to submit, view and update requests.
Power Automate	Automates tasks between apps using triggers and actions (called flows).	Sends notifications , routes a basic approval and sends reminders .
Power BI	Connects to data, prepares it and presents interactive reports and dashboards.	Shows request volume, status, priority and trends ; distributed via a Power BI App .



Power BI App vs Power Apps: two different products

These names sound similar but serve very different purposes. Keep this distinction in mind throughout the programme.

Aspect	Power BI App	Power Apps
Purpose	Packages and distributes finished Power BI reports and dashboards to an audience.	Builds business applications that capture and update data.
What you create	A read-only collection of reports, with navigation and sections.	Screens, forms, buttons and galleries (a working application).
Where it is built	Power BI Service, from a workspace (Create app).	Power Apps Studio (make.powerapps.com).

Aspect	Power BI App	Power Apps
What users do	View, filter and explore reports.	Enter, edit and submit information.
In our case	Managers open the “PC MTBE Improvement Insights” app to view request reports.	Staff use the “Request App” to submit and update improvement requests.

The training case: PC MTBE Process Improvement Request

Imagine that PC MTBE wants to collect ideas for improving work processes — for example, automating a manual log, digitalising a paper form or improving a monthly report. Today these ideas arrive by email and are tracked in a spreadsheet. You will replace this with a connected digital process:

1. An employee submits a request through the **Power Apps** Canvas App.
2. The request is saved as a record in a **SharePoint List**.
3. **Power Automate** assigns a Request ID, notifies the team and sends the request for approval.
4. The approver responds in Outlook or Teams; the list status is updated automatically. Pending requests trigger a daily reminder.
5. **Power BI** analyses status, priority and trends, and the report is shared with managers through a **Power BI App**.

FICTIONAL TRAINING DATA

All names, departments, dates and requests used in this manual are **fictional training data** created for learning purposes. They do not represent actual PC MTBE information, staff or performance.

Key terms you will hear today

Term	Plain-language meaning
Microsoft 365 tenant	Your organisation's private space in Microsoft's cloud, managed by your IT administrators.
Connector	A ready-made link that lets Power Apps, Power Automate or Power BI talk to a service such as SharePoint or Outlook.
Environment	A container in Power Platform that holds apps and flows. Your organisation may have separate environments for training, testing and production.
Low-code	Building solutions mainly through visual designers and short formulas, similar to Excel formulas.

MY NOTES

SECTION 2

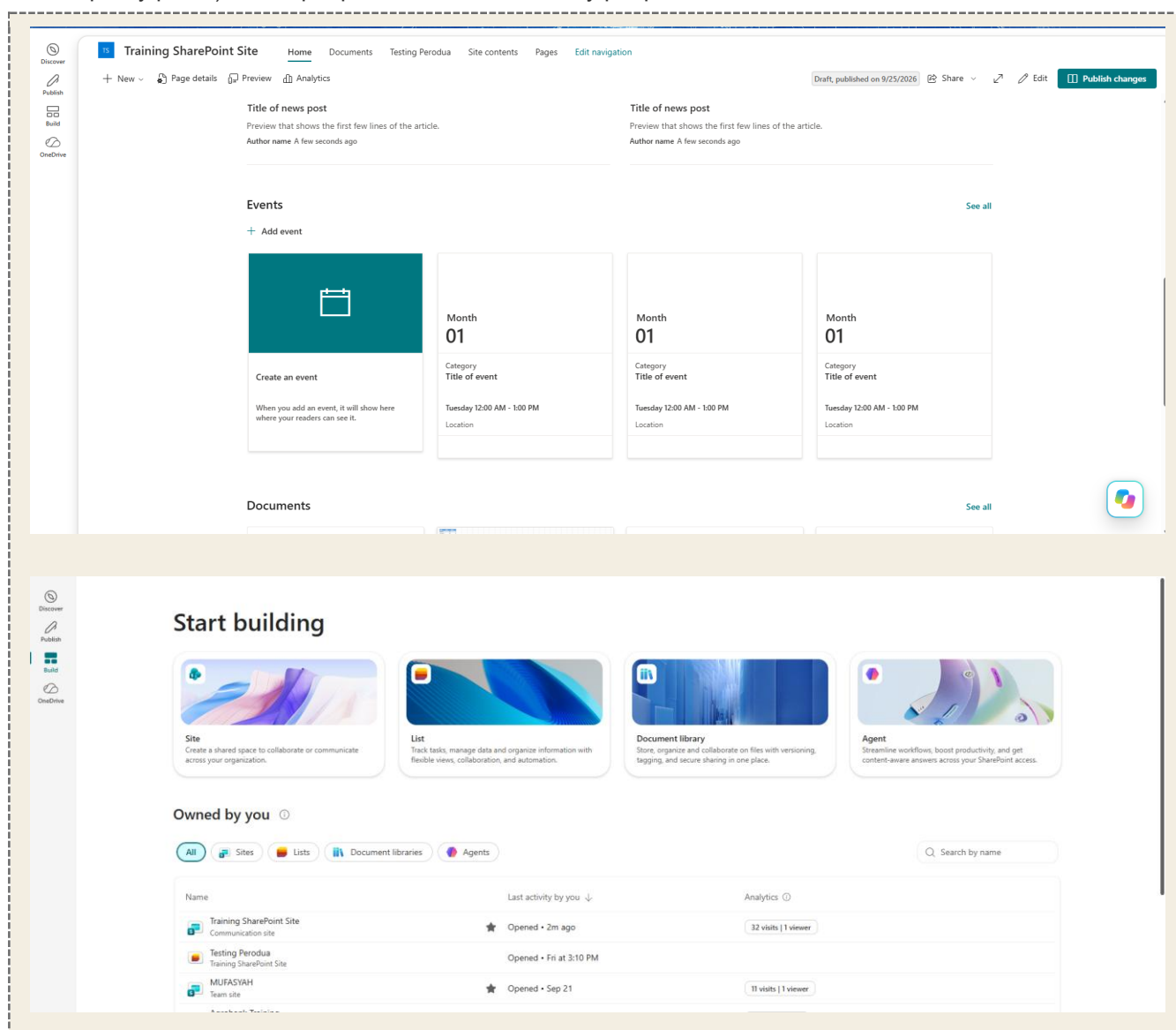
SharePoint fundamentals

Where your team's documents and lists live — securely, with history, and accessible from anywhere.

2.1 SharePoint sites

A **SharePoint site** is a web-based workspace for a team, project or department. Each site contains pages, document libraries and lists, and has its own members and permissions. There are two common site types:

- **Team site** — for collaboration among a working group (for example, the HSE team). Usually connected to a Microsoft 365 Group and often to a Microsoft Teams team, so files in a Teams channel are actually stored in SharePoint.
- **Communication site** — for publishing information to a wider audience (for example, a company news or policy portal). A few people create content; many people read it.



2.2 Document libraries and collaboration

A **document library** is a special folder on a site that stores files such as Word, Excel, PowerPoint and PDF documents. Compared with a shared network drive, a library offers:

- **Co-authoring** — several people can edit the same Word, Excel or PowerPoint file at the same time; you will see their names and cursors.
- **Sharing links** — share a link instead of attaching a copy to email, so everyone works on one version.
- **Metadata** — extra columns (for example, Department or Document Type) that let you filter and group files instead of relying on deep folders.
- **Version history** — every save creates a version that you can view or restore.

Try it: view and restore a previous version

1. Open the **Documents** library and upload or create a Word file named **HSE Toolbox Talk.docx**.
2. Open it in the browser, add one sentence, and close it. Repeat once more with a different sentence.
3. In the library, select the file, click the ... (Show actions) menu and choose **Version history**.
4. Select a date to open that version, or use the drop-down beside it and choose **Restore** to make it the current version.

2.3 Permissions at a beginner level

Permissions control who can see or change content. Most sites use three standard groups:

Group	Typical permission level	Can do
Site Owners	Full Control	Manage settings, members and permissions; create lists and libraries.
Site Members	Edit	Add, edit and delete documents and list items.
Site Visitors	Read	View and download content only.

By default, libraries and lists **inherit** permissions from the site. Avoid giving permissions to individual files unless there is a clear reason — it quickly becomes difficult to manage. When sharing, prefer **People in your organisation** or **Specific people** links over broader options.

CHECK WITH YOUR ADMINISTRATOR

Your organisation may restrict who can create sites, lists and external sharing links. If you cannot see **New > List** or the **Share** options described here, ask your SharePoint administrator or site owner for the correct permission on the training site.

2.4 Good practices for organising digital documents

- **Use clear, consistent file names:** for example, **Dept_Topic_YYYY-MM-DD** such as **HSE_PermitToWork_2026-09-01.docx**. Avoid “final” or “v2” — version history already tracks versions.
- **Keep folder structures shallow** (no more than two or three levels) and use metadata columns for filtering.
- **Store once, share links** rather than emailing attachments.
- **Separate working and final documents**, for example with a Status column (Draft, Under Review, Approved).
- **Review access regularly** and remove people who no longer need it.

2.5 SharePoint Lists

A **SharePoint List** is like a smart, shared table. Each row is an **item** and each column has a **type** that controls what can be entered. Lists are ideal for tracking requests, issues, assets or registrations, and they are an excellent beginner data source for Power Apps, Power Automate and Power BI.

Column type	Use it for	Example in our list
Single line of text	Short text up to 255 characters	Request Title, Request ID, Owner
Multiple lines of text	Longer descriptions	Description
Choice	A fixed set of options (reduces typing errors)	Department, Request Type, Priority, Status
Date and time	Dates, with or without time	Submission Date, Completion Date
Person	People from your organisation directory	Created By (built in)

HANDS-ON EXERCISE 2.1

Create the Process Improvement Requests list

Objective	Create the SharePoint List that stores all requests in the case study, using suitable column types.
Estimated duration	25 minutes
Prerequisites	Access to the training SharePoint site with Edit or Owner permission. Your trainer will give you the site URL.

Steps

1. Open the training site. Select **+ New > List**, then choose **Blank list**.
2. Name the list **Process Improvement Requests** (add your initials if several participants share one site, e.g. **Process Improvement Requests - FS**). Add the description "Training list – fictional data" and click **Create**.
3. Rename the built-in **Title** column: click the **Title** column heading > **Column settings > Edit**, change the name to **Request Title** and click **Save**. (The internal name remains "Title"; this is normal.)
4. Select **+ Add column** and create each column in the table below, in the order shown. For Choice columns, type each option on its own line.
5. For **Status**, set the **Default value** to **Submitted**. For **Submission Date**, expand **More options** and set the default value to **Today's date**.
6. Check that all columns appear in the list and that the column types are correct.

Column name	Type	Settings / choices
Request Title	Single line of text	Built-in Title column (renamed); required
Request ID	Single line of text	Not required – filled automatically by Power Automate in Section 4
Submission Date	Date and time	Date only; default = Today's date
Department	Choice	Operations, Maintenance, Finance, Human Resources, Procurement, HSE, IT
Request Type	Choice	Process Automation, Reporting Improvement, Document Management, Digital Form, Other
Description	Multiple lines of text	Plain text

Column name	Type	Settings / choices
Priority	Choice	Low, Medium, High
Status	Choice	Submitted, Pending Approval, Approved, Rejected, In Progress, Completed; default = Submitted
Owner	Single line of text	Name of the person responsible for implementation
Completion Date	Date and time	Date only; not required

NOTE

In a live solution, **Owner** would normally be a **Person** column linked to your directory. We use text in this course because the sample owners are fictional and do not exist in the directory.

Expected result

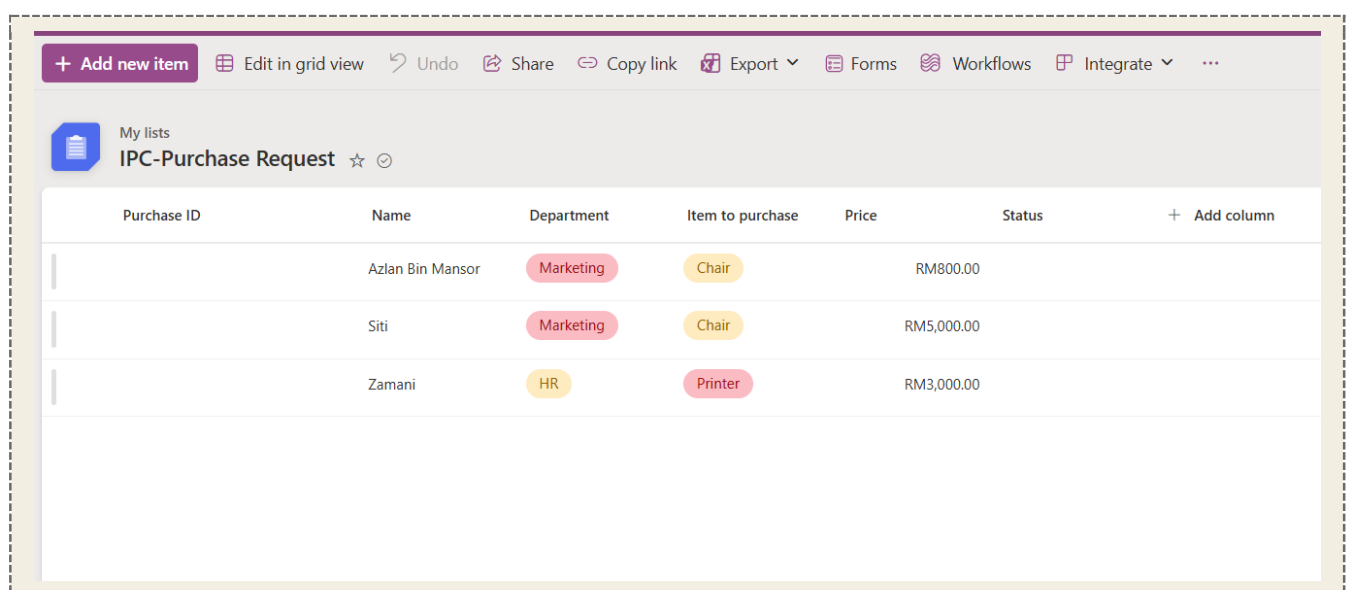
An empty list named Process Improvement Requests with ten columns: Request Title, Request ID, Submission Date, Department, Request Type, Description, Priority, Status, Owner and Completion Date.

Checkpoint

- ☐ The list is created on the training site and its name is exactly as instructed.
- ☐ Department, Request Type, Priority and Status are **Choice** columns with the correct options.
- ☐ Status defaults to **Submitted** and Submission Date defaults to today.

Common errors and how to resolve them

Common error	How to resolve it
+ New > List is missing or greyed out.	You probably have Read permission only. Ask the site owner to add you to Site Members or Site Owners.
A Choice option was misspelt (e.g. "Compleeted").	Open Column settings > Edit , correct the option and Save . Update any items that used the old value.



HANDS-ON EXERCISE 2.2

Add fictional data and create useful views

Objective	Populate the list with the sample dataset and create views that help people find the right requests quickly.
Estimated duration	20 minutes
Prerequisites	Exercise 2.1 completed.

Steps

1. Select **Edit in grid view** at the top of the list. The list now behaves like a spreadsheet.
2. Enter the twelve fictional records from the sample dataset below. You can copy rows from the trainer's Excel file and paste them into grid view. Leave **Request ID** blank for now if your trainer prefers to generate IDs with Power Automate; otherwise, enter the IDs shown.
3. Select **Exit grid view**.
4. Create a grouped view: open the **Status** column menu > **Group by Status**. Then open the view menu (for example, **All Items**) > **Save view as**, name it **By Status** and save.
5. Create a filtered view: return to **All Items**, open the **Priority** column menu > **Filter by** > tick **High**. Then filter **Status** to show everything except **Completed** and **Rejected**. Save the view as **Open High Priority**.
6. Switch between views and confirm that each shows the expected records.

Sample dataset — fictional training data

Request ID	Submitted	Department	Request Type	Request Title	Priority	Status	Owner	Completed
REQ-0001	02/06/2026	Operations	Process Automation	Automate daily shift handover log	High	Completed	Aina Rahman	30/06/2026
REQ-0002	09/06/2026	Finance	Reporting Improvement	Monthly cost report dashboard	Medium	Completed	Daniel Lee	21/07/2026
REQ-0003	16/06/2026	HSE	Digital Form	Digital permit-to-work checklist	High	In Progress	Nurul Huda	
REQ-0004	23/06/2026	Human Resources	Document Management	Central library for HR policies	Low	Completed	Kavitha Raj	14/07/2026
REQ-0005	07/07/2026	Maintenance	Process Automation	Spare parts request notification	Medium	Approved	Hafiz Ismail	
REQ-0006	14/07/2026	Procurement	Digital Form	Vendor registration form	Medium	Pending Approval	Daniel Lee	
REQ-0007	28/07/2026	Operations	Reporting Improvement	Weekly production KPI report	High	In Progress	Aina Rahman	
REQ-0008	04/08/2026	IT	Process Automation	Laptop request approval flow	Low	Completed	Faris Hakimi	25/08/2026
REQ-0009	18/08/2026	Finance	Process Automation	Invoice approval reminder	High	Rejected	Kavitha Raj	
REQ-0010	01/09/2026	HSE	Reporting Improvement	Incident trend dashboard	High	Submitted	Nurul Huda	
REQ-0011	08/09/2026	Maintenance	Document Management	Equipment manual library	Medium	Submitted	Hafiz Ismail	
REQ-0012	15/09/2026	Human Resources	Digital Form	Training nomination form	Low	Pending Approval	Faris Hakimi	

Descriptions (enter one sentence per record, for example):

- **REQ-0001:** Replace the paper shift handover book with a digital log that supervisors can search.
- **REQ-0003:** Convert the paper permit-to-work checklist into a mobile form with photo attachments.

- **REQ-0006:** Allow vendors' registration details to be captured once and routed for review automatically.
- **REQ-0010:** Visualise monthly incident and near-miss trends by area for the HSE committee.
- For the other records, write a short description of at least 20 characters in your own words.

Expected result

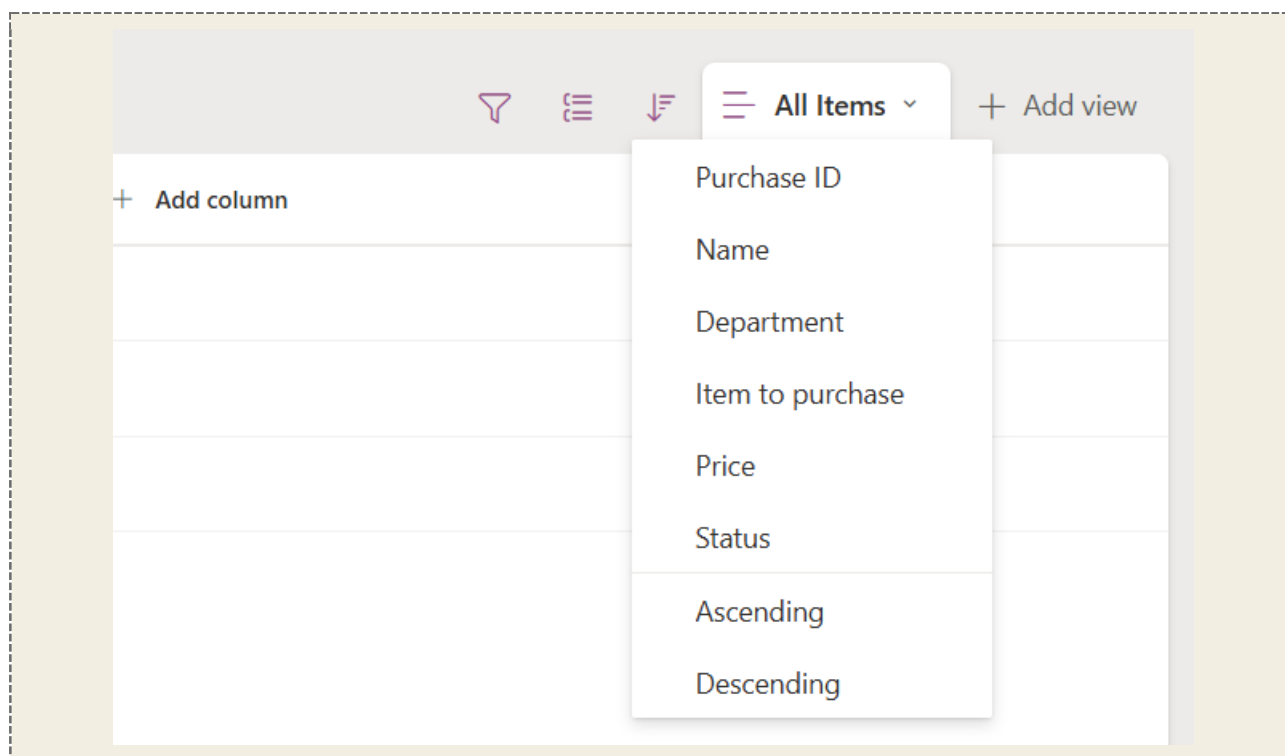
The list contains twelve fictional records. The By Status view groups items by status, and the Open High Priority view shows REQ-0003, REQ-0007 and REQ-0010.

Checkpoint

- ☐ Twelve items appear in **All Items**.
- ☐ The **By Status** view groups records under each status.
- ☐ The **Open High Priority** view shows exactly three records.

Common errors and how to resolve them

Common error	How to resolve it
Pasting from Excel puts data in the wrong columns.	Make sure the column order in grid view matches your Excel columns, or paste one column at a time. Dates must use a valid format for your regional settings.
A choice value is rejected or appears blank.	The pasted text must exactly match a choice option (including spacing and capital letters).



MY NOTES

SECTION 3

Power Apps fundamentals

Build a simple business app on top of your SharePoint List — no traditional coding required.

3.1 What is Power Apps?

Power Apps lets you build business applications that run in a web browser, in Microsoft Teams and on the Power Apps mobile app. There are two main types:

- **Canvas App** — you design the screens freely, like arranging objects on a PowerPoint slide, and connect them to data. Best for task-focused apps such as forms, checklists and request trackers. **This programme uses Canvas Apps.**
- **Model-driven App** — the layout is generated from a structured data model in Microsoft Dataverse. Best for complex business processes; usually requires additional licensing.

You build Canvas Apps in **Power Apps Studio**, opened from make.powerapps.com. Apps are saved in an **environment**, and each app uses **connectors** to reach data sources.

3.2 How a Canvas App connects to a SharePoint List

When you add the SharePoint list as a **data source**, the app can read, create and update list items. The app does not keep its own copy of the data: every submission is written straight to the list, which is why the same data then appears in Power Automate and Power BI.

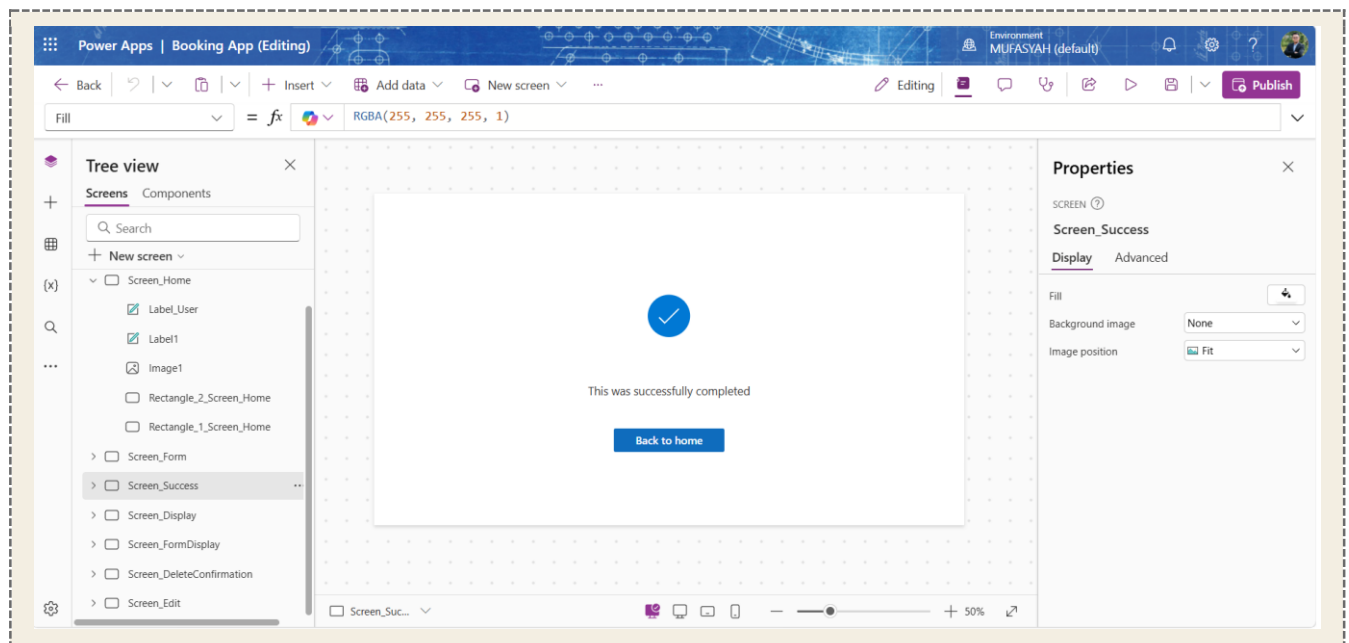
- A **Gallery** displays many records (the list of requests).
- A **Display form** shows one record in read-only mode.
- An **Edit form** creates a new record or updates an existing one.
- **Delegation** means the app asks SharePoint to filter data on the server instead of downloading everything. If you see a yellow warning triangle about delegation, the formula may only work on the first 500 records. For a training list this is not a problem, but it matters for large lists.

CHECK WITH YOUR ADMINISTRATOR

Canvas Apps that use standard connectors such as SharePoint are generally covered by Microsoft 365 licences. Premium connectors (for example, SQL Server or Dataverse) require Power Apps licences. Your organisation also decides which **environment** you should build in and who may share apps. Confirm both before building production apps.

3.3 A quick tour of Power Apps Studio

Area	What it is for
Tree view (left)	Lists all screens and controls. Use it to select and rename controls.
Canvas (centre)	The design area where you arrange controls on each screen.
Property drop-down and formula bar (top)	Choose a property (for example, Text , Items , OnSelect) and write its formula.
Properties pane (right)	Change appearance and settings without formulas.
Data (left toolbar)	Add or refresh data sources such as the SharePoint list.
Preview (▶ or F5)	Run the app to test it. Press Esc to return to editing.



3.4 Essential controls and Power Fx formulas

Power Fx is the formula language of Power Apps. If you have written an Excel formula, you already know the idea: a property is set to a formula that is recalculated automatically.

Control or function	What it does	Example
Gallery	Shows a scrollable list of records	<code>Items = the SharePoint list</code>
Text input / Combo box / Date picker	Let users type, choose or pick values	Description, Priority, Submission Date
Button / Icon	Runs an action when selected	<code>OnSelect = SubmitForm(EditForm1)</code>
Navigate / Back	Move between screens	<code>Navigate(EditScreen1)</code>
NewForm / EditForm / SubmitForm	Prepare a form for a new record, edit a record, or save it	<code>NewForm(EditForm1)</code>
Filter / SortByColumns / StartsWith	Narrow and order records	<code>Filter(List, Status.Value = "Submitted")</code>
If / IsBlank / Len / Notify	Check conditions and show messages	<code>If(IsBlank(txtTitle.Text), Notify("Required"))</code>

HANDS-ON EXERCISE 3.1

Generate a Canvas App from the SharePoint List

Objective	Create a working three-screen app that views, adds and edits requests, generated automatically from the list.
Estimated duration	20 minutes
Prerequisites	Exercises 2.1 and 2.2 completed; access to make.powerapps.com in the environment specified by your trainer.

Steps

1. Go to make.powerapps.com and confirm the correct **Environment** is selected at the top right.
2. On the home page, choose the option to create an app from **SharePoint** data. Depending on your version, this appears as **Create > SharePoint, Start with data > Connect to external data > SharePoint**, or similar.
3. Select or create a SharePoint connection, enter the training site URL (or choose it from the recent sites list) and click **Connect**.
4. Select the **Process Improvement Requests** list and click **Connect**. Wait while Power Apps generates the app.
5. Review the three generated screens in the Tree view: **BrowseScreen1** (gallery of requests), **DetailScreen1** (read-only details) and **EditScreen1** (form for new or edited requests).
6. Press **F5** (Preview). Select a request to open its details, click the pencil icon to edit it, and use the **+** icon to open a blank form. Press **Esc** to exit.
7. Select **File > Save As** (or the Save icon), name the app **PC MTBE Request App - [Your Initials]** and save.

Expected result

A saved Canvas App with three screens that displays the twelve fictional requests and allows you to open, edit and create items.

Checkpoint

- ☐ The gallery shows your fictional records.
- ☐ Selecting a record opens DetailScreen1 with the correct values.
- ☐ The app is saved with your initials in the name.

Common errors and how to resolve them

Common error	How to resolve it
The list does not appear in the list picker.	Check that you entered the site URL (e.g. <code>.../sites/Training</code>), not the list URL, and that you can open the list in SharePoint.
"We couldn't generate the app" or a connection error appears.	Refresh the page, sign in again and recreate the SharePoint connection. If it persists, the connector may be blocked by a data policy — check with your administrator.

HANDS-ON EXERCISE 3.2

Customise, validate and test the app

Objective	Make the app easier to use by improving the gallery, setting default values and preventing incomplete submissions.
Estimated duration	60 minutes (including peer testing)
Prerequisites	Exercise 3.1 completed and the app open in Power Apps Studio.

Steps

1. **Rename the title:** on BrowseScreen1 select the title label and set its **Text** property to **"Process Improvement Requests"**.
2. **Show useful fields in the gallery:** select the first item in the gallery and set the three labels to show **ThisItem.Title**, **ThisItem.Department.Value** and **ThisItem.Status.Value**.
3. **Show newest requests first:** select the gallery (BrowseGallery1) and set its **Items** property to:

```
SortByColumns(
  Filter(
    'Process Improvement Requests',
    StartsWith(Title, TextSearchBox1.Text)
  ),
  "ID",
  SortOrder.Descending
)
```

4. **Colour-code the status:** select the status label in the gallery and set its **Color** property to:

```
Switch(
  ThisItem.Status.Value,
  "Completed", RGBA(74, 90, 43, 1),
  "Rejected", RGBA(181, 84, 58, 1),
  "Pending Approval", RGBA(191, 128, 20, 1),
  RGBA(46, 46, 46, 1)
)
```

5. **Arrange the form:** on EditScreen1 select **EditForm1**, open **Edit fields** in the Properties pane, and drag fields into this order: Request Title, Department, Request Type, Priority, Description, Submission Date, Status, Owner, Completion Date, Request ID. Set **Columns** to 1 for mobile-friendly layout.
6. **Rename key controls** in the Tree view (double-click a name): the Request Title text box to **txtTitle**, the Description text box to **txtDescription** and the Priority combo box to **cmbPriority**. Clear names make formulas easier to read.
7. **Set defaults for new requests:** select the **Status** data card, unlock it (**Advanced > Unlock to change properties**) and set its **Default** property to:

```
If(EditForm1.Mode = FormMode.New, {Value: "Submitted"}, ThisItem.Status)
```

8. Unlock the **Submission Date** card and set its **Default** to:

```
If(EditForm1.Mode = FormMode.New, Today(), ThisItem.'Submission Date')
```

9. **Protect system fields:** set the **DisplayMode** of the Request ID card to **DisplayMode.View**, because Power Automate will fill it in.

10. **Add validation:** select the save (tick) icon on EditScreen1 and replace its **OnSelect** formula with:

```
If(
  IsBlank(txtTitle.Text) || IsEmpty(cmbPriority.SelectedItems)
  || Len(txtDescription.Text) < 20,
  Notify("Please enter a title, choose a priority and write a description
    of at least 20 characters.", NotificationType.Error),
  SubmitForm(EditForm1)
)
```

11. **Confirm success:** select **EditForm1** and set **OnSuccess** to:

```
Notify("Your request has been saved.", NotificationType.Success);
Back()
```

12. **Test** using the scenarios in the table below. Then **Save** and **Publish** the app (**File > Publish**, or the Publish icon).

Test scenarios

#	Scenario	Action	Expected result
1	Missing information	Click + and save with an empty title	Red error message; nothing is saved
2	Short description	Enter "Fix form" as the description	Red error message
3	Valid new request	Complete all required fields and save	Green message; new item appears at the top of the gallery with status Submitted
4	Update a request	Open REQ-0005, edit it and change Status to In Progress	Change visible in the app and in the SharePoint list

Expected result

A published app that lists newest requests first, colour-codes status, pre-fills today's date and Submitted status for new requests, and blocks incomplete submissions with a clear message.

Checkpoint

- ☐ All four test scenarios behave as expected.
- ☐ A request created in the app appears in the SharePoint list.
- ☐ The app is saved **and** published.

Common errors and how to resolve them

Common error	How to resolve it
Red error underline: "Name isn't valid. 'txtDescription' isn't recognised."	The control name in your formula does not match the Tree view. Rename the control or update the formula so the names match exactly.
Changes work in Studio but colleagues still see the old version.	You saved but did not publish. Use Publish , then ask users to close and reopen the app.

CHECK WITH YOUR ADMINISTRATOR

To let colleagues use your app, select **Share** from the app menu and add people or a security group; they also need at least **Edit** permission on the SharePoint list to submit requests. Your organisation may require apps to be moved to a production environment before sharing — check with your administrator.

3.5 Common Power Apps issues

Symptom	Likely cause	What to do
New list column does not appear in the form	The data source has not been refreshed	Open Data , select the list's ... > Refresh , then add the field via Edit fields
Yellow delegation warning	A function cannot be processed by SharePoint on the server	Acceptable for small lists; for large lists, use delegable functions such as StartsWith and Filter with simple conditions
"The data source may not support this operation"	Missing permission on the list	Ask the site owner for Edit permission

Symptom	Likely cause	What to do
App is slow to load	Too many controls or data calls on the first screen	Keep screens simple; load only the data you need

MY NOTES

SECTION 4

Power Automate fundamentals

Let the system send the emails, chase the approvals and update the status for you.

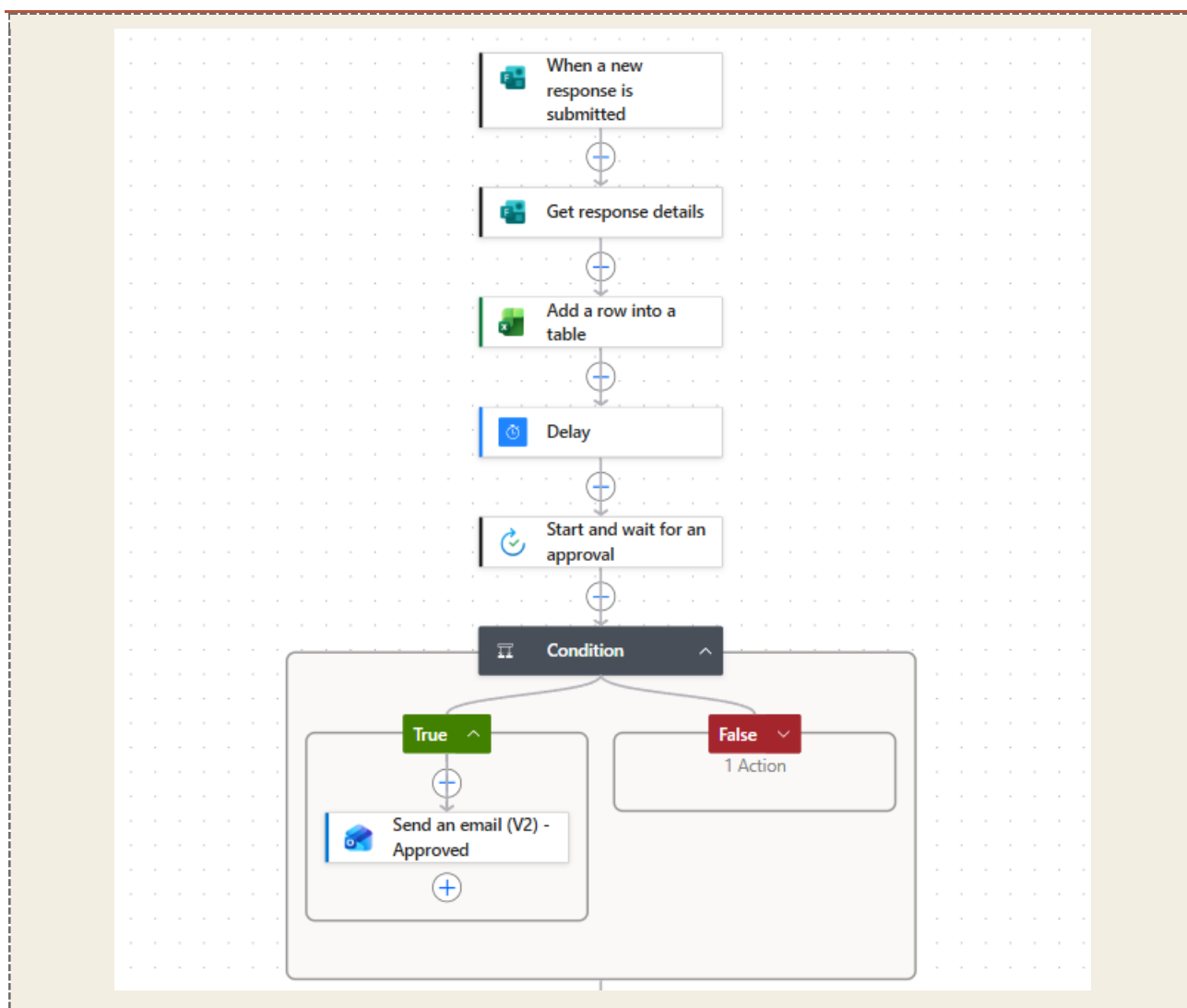
4.1 What is Power Automate?

Power Automate creates **flows** — automated sequences of steps that run between apps and services. A flow can send emails, request approvals, copy files, post Teams messages or update SharePoint items, without anyone having to remember to do it.

Flow type	Starts when...	Example in our case
Automated cloud flow	Something happens (an event)	A new request is created in the list
Instant cloud flow	Someone presses a button	A manager manually sends a summary from the app or Teams
Scheduled cloud flow	A set time arrives	Every weekday at 8:00 AM, check for pending approvals
Desktop flow	Automating legacy desktop applications (RPA)	Not covered in this programme

4.2 The building blocks of a flow

- **Trigger** — the event that starts the flow. Every flow has exactly one trigger (for example, **When an item is created**).
- **Action** — a task the flow performs (for example, **Send an email (V2)** or **Update item**). A flow can have many actions.
- **Dynamic content** — values produced by earlier steps, such as the request title or the creator's email, that you insert into later steps.
- **Condition** — a yes/no decision that sends the flow down one of two branches (for example, approved or rejected).
- **Expression** — a small formula for calculations or formatting, such as `addDays(utcNow(), -3)`.
- **Run history** — a record of every time the flow ran, showing which steps succeeded or failed and why.



CHECK WITH YOUR ADMINISTRATOR

The SharePoint, Office 365 Outlook and Approvals connectors are **standard** connectors included with most Microsoft 365 licences. **Premium** connectors (marked with a diamond icon) require a Power Automate licence. Your organisation may also apply **data loss prevention (DLP) policies** that block certain connector combinations. If a connector is unavailable or a flow is suspended, check with your administrator.

HANDS-ON EXERCISE 4.1

New-request notification and Request ID

Objective	Automatically assign a Request ID and send an email notification whenever a new request is created.
Estimated duration	25 minutes
Prerequisites	The SharePoint list and the Power Apps app from Sections 2 and 3; access to make.powerautomate.com .

Steps

1. Go to make.powerautomate.com and check the **Environment** is the same one used for Power Apps.

2. Select **Create > Automated cloud flow**. Name it **PIR - New Request - [Your Initials]**, search for the trigger **When an item is created** (SharePoint) and click **Create**.
3. In the trigger, select the training site in **Site Address** and **Process Improvement Requests** in **List Name**.
4. Add a new step: **SharePoint > Update item**. Select the same site and list, then set:
 - **Id**: dynamic content **ID** (from the trigger).
 - **Request Title**: dynamic content **Title** (required by the action, so the existing title is kept).
 - **Request ID**: open the expression editor (fx) and enter the expression below.
5. Request ID expression:


```
concat('REQ-', formatNumber(triggerOutputs()?['body/ID'], '0000'))
```
6. Add **Office 365 Outlook > Send an email (V2)** and set:
 - **To**: your own email address (representing the Process Improvement team).
 - **Subject**: **New improvement request**: followed by dynamic content **Title**.
 - **Body**: include Department, Request Type, Priority, Description and **Created By DisplayName**, plus the link **Link to item**.
7. Select **Save**, then **Test > Manually > Test**. Submit a new request from your Power Apps app.
8. Return to the flow and watch the run. Every step should show a green tick. Check your inbox and the SharePoint list.

Expected result

Each new request receives an ID such as REQ-0013 within a minute, and an email summarising the request arrives in your inbox.

Checkpoint

- ☐ The flow run shows green ticks for all steps.
- ☐ The new item in the list has a Request ID in the REQ-0000 format.
- ☐ The email contains the correct title, priority and a working link.

Common errors and how to resolve them

Common error	How to resolve it
The flow does not start after submitting a request.	Check the trigger points to the correct site and list. Triggers can take up to a few minutes; use Test mode so you can see the run immediately.
"The expression is invalid" when adding Request ID.	Copy the expression exactly, including single quotes and brackets, into the Expression tab — not the text box itself.

HANDS-ON EXERCISE 4.2

Basic approval workflow

Objective	Route each new request to an approver and update its status automatically based on the decision.
Estimated duration	35 minutes
Prerequisites	Exercise 4.1 completed and working.

Steps

1. Open the flow from Exercise 4.1 and select **Edit**. In the existing **Update item** action, also set **Status Value** to **Pending Approval**.
2. After **Send an email (V2)**, add **Approvals > Start and wait for an approval** and set:

- **Approval type:** Approve/Reject – First to respond.
 - **Title:** **Approval needed:** followed by **Title**.
 - **Assigned to:** your own email address (in real life, the department head or process owner).
 - **Details:** Department, Request Type, Priority and Description; **Item link:** **Link to item**.
3. Add a **Condition** below the approval. Set the left value to dynamic content **Outcome**, choose **is equal to**, and type **Approve** on the right.
 4. In **If yes**, add **Update item** (same site and list): **Id** = ID, **Request Title** = Title, **Status Value** = **Approved**. Then add **Send an email (V2)** to **Created By Email** with the subject **Your request has been approved**.
 5. In **If no**, add **Update item** with **Status Value** = **Rejected**, and **Send an email (V2)** to **Created By Email**. In the body, add the dynamic content **Responses Comments** so the requester sees the reason. Power Automate may wrap this in an **Apply to each** loop — this is normal.
 6. **Save** and **Test** by submitting a new request. Open the approval in Outlook, Teams (Approvals app) or make.powerautomate.com > **Approvals**, add a comment and choose **Approve**.
 7. Test again with **Reject**. Confirm the list status and the requester email for both cases.

Expected result

New requests move to Pending Approval, the approver receives an actionable approval, and the list status changes to Approved or Rejected with an email sent back to the requester.

Checkpoint

- ☐ The approval request arrived with the correct details.
- ☐ Approve and Reject paths both updated the status correctly.
- ☐ The rejection email includes the approver's comment.

Common errors and how to resolve them

Common error	How to resolve it
The condition always goes to If no .	Check the right-hand value is exactly Approve (not "Approved"). The Outcome value is case-sensitive.
The flow shows "Running" for a long time.	This is expected — it is waiting for the approver. Approvals can wait up to 30 days before the flow times out.

HANDS-ON EXERCISE 4.3

Scheduled reminder for pending approvals

Objective	Send a daily reminder listing requests that have been waiting for approval for more than three days.
Estimated duration	25 minutes
Prerequisites	Exercise 4.2 completed; at least one list item with Status = Pending Approval and a Submission Date more than three days ago (REQ-0006 and REQ-0012 in the sample data).

Steps

1. Select **Create > Scheduled cloud flow**. Name it **PIR - Daily Reminder - [Your Initials]**. Set it to repeat every **1 Day**, then click **Create**.
2. Open the **Recurrence** trigger, expand advanced options and set **Time zone** to **(UTC+08:00) Kuala Lumpur, Singapore**, **At these hours** to **8** and **At these minutes** to **0**.

3. Add **SharePoint > Get items**, select the site and list, and in **Filter Query** enter:

```
Status eq 'Pending Approval'
```

4. Add **Control > Apply to each** and choose **value** (from Get items). Inside it, add a **Condition**: left = dynamic content **Submission Date**, operator **is less than or equal to**, right = the expression:

```
addDays(utcNow(), -3, 'yyyy-MM-dd')
```

5. In **If yes**, add **Send an email (V2)** to your own address. Subject: **Reminder: approval pending for** followed by **Request ID**. Body: include Request Title, Department, Priority and Submission Date.
6. **Save**, then **Test > Manually**. You do not need to wait for 8:00 AM.

Expected result

One reminder email for each request that has been pending approval for more than three days (for example, REQ-0006 and REQ-0012 in the sample data).

Checkpoint

- ☐ The flow runs successfully when tested manually.
- ☐ Only requests with Pending Approval status and an old submission date generate reminders.
- ☐ The recurrence uses the Kuala Lumpur time zone.

Common errors and how to resolve them

Common error	How to resolve it
Get items fails with "Column ... does not exist".	Filter Query uses internal column names. Status is fine; for other columns, check the internal name in the URL of List settings > column.
No emails are sent although items are pending.	Check the Submission Date values are more than three days ago, and that the right side of the condition is an expression , not typed text.

4.3 Testing, monitoring and troubleshooting flows

- **Test before you share**: use **Test > Manually** to watch each step run in real time.
- **Read the run history**: open the flow's details page, select a failed run, and expand the step with the red exclamation mark to read the error message.
- **Rename actions** (for example, "Update status to Approved") so the flow is easy to understand later.
- **Avoid loops**: a flow triggered by **When an item is created or modified** that also updates the same item can trigger itself repeatedly. We use **When an item is created** to avoid this.
- **Ownership**: flows run with the owner's connections. Add a co-owner so the flow keeps working if you change roles.

MY NOTES

SECTION 5

Power BI fundamentals and the Power BI App

Turn request data into clear insight — and deliver it to managers in one place.

5.1 The Power BI ecosystem

Power BI is Microsoft's business intelligence platform. You typically build reports in **Power BI Desktop** (a free Windows application) and share them through the **Power BI Service** (app.powerbi.com).

Item	What it is
Semantic model (dataset)	The prepared data and calculations behind a report.
Report	One or more interactive pages of visuals built on a semantic model.
Dashboard	A single page in the Power BI Service made of tiles pinned from one or more reports.
Workspace	A shared area in the Power BI Service where a team creates and manages content.
Power BI App	A packaged, read-only bundle of reports and dashboards from one workspace, published to a defined audience.

The typical workflow is: **Get data** → **Transform (Power Query)** → **Model and measures** → **Visualise** → **Publish** → **Share through an App**.

5.2 Preparing data with Power Query

Power Query is the data preparation tool inside Power BI (and Excel). Each change you make is recorded as an **Applied Step**, so the same cleaning is repeated automatically every time the data refreshes. Typical tasks for our list include removing system columns, renaming columns, setting correct data types and extracting values from Choice columns.

5.3 Measures and DAX in brief

A **measure** is a reusable calculation written in **DAX** (Data Analysis Expressions), Power BI's formula language. Measures recalculate automatically when users click a slicer or visual. You will create these five measures:

```
Total Requests = COUNTROWS(Requests)

Completed Requests =
    CALCULATE([Total Requests], Requests[Status] = "Completed")

Open Requests =
    CALCULATE([Total Requests],
        NOT Requests[Status] IN {"Completed", "Rejected"})

Completion Rate = DIVIDE([Completed Requests], [Total Requests])

Avg Days to Complete =
    AVERAGEX(
        FILTER(Requests, NOT ISBLANK(Requests[Completion Date])),
        DATEDIFF(Requests[Submission Date], Requests[Completion Date], DAY)
    )
```

5.4 Essential visuals and design principles

Visual	Best for	In our report
Card / KPI card	One important number	Total, Open, Completion Rate, Avg Days
Clustered bar chart	Comparing categories	Requests by Department
Donut chart	Parts of a whole (few categories)	Requests by Status
Stacked column chart	Categories split by a second field	Priority by Request Type
Line chart	Trends over time	Requests submitted per month
Slicer	Letting users filter the page	Department, Priority, Submission Date
Table	Detail for follow-up	Open requests with owner and age

- **Start with the question:** “How many requests are open, where are they, and are we improving?” Every visual should help answer it.
- **Put key numbers top-left:** readers scan from top-left to bottom-right, so place KPI cards first and detail tables last.
- **Limit each page to about six to eight visuals** and align them to a grid.
- **Use colour with purpose:** one neutral colour for most data and one accent colour for what needs attention (for example, High priority).
- **Write titles that explain:** “Open requests by department” is clearer than “Count of ID by Department”.

HANDS-ON EXERCISE 5.1

Connect to and prepare the request data

Objective	Import the SharePoint list into Power BI Desktop and prepare clean, correctly typed data.
Estimated duration	30 minutes
Prerequisites	Power BI Desktop installed; the list contains the twelve fictional records and any requests you added.

Steps

1. Open **Power BI Desktop** and sign in with your work account (top right).
2. Select **Home > Get data > More...**, search for **SharePoint Online list** and click **Connect**.
3. Enter the **site URL** only (for example, [https://\[tenant\].sharepoint.com/sites/Training](https://[tenant].sharepoint.com/sites/Training)). Select **2.0** as the implementation if offered, then **OK**. Sign in with **Microsoft account** if prompted.
4. In the Navigator, tick **Process Improvement Requests** and click **Transform Data** (not Load).
5. In Power Query, rename the query to **Requests** (Query Settings > Name).
6. Select **Home > Choose Columns** and keep only: ID, Title, Request ID, Submission Date, Department, Request Type, Description, Priority, Status, Owner, Completion Date, Created.
7. If Choice columns show **Record** instead of text, click the expand icon in the column header and select **Value** (untick “Use original column name as prefix”).
8. Rename **Title** to **Request Title**. Set data types: **Submission Date** and **Completion Date** = Date; **ID** = Whole number; text columns = Text.
9. Add a month column: select **Submission Date**, then **Add Column > Date > Month > Start of Month**, and rename it **Submission Month**.
10. Select **Home > Close & Apply**.

Expected result

A table named **Requests** appears in the Data pane with clean column names and correct data types, including Submission Month.

Checkpoint

- ☐ The **Requests** table contains all list items.
- ☐ Department, Priority and Status show text values, not “Record”.
- ☐ Both date columns have the calendar icon (Date type).

Common errors and how to resolve them

Common error	How to resolve it
“Access to the resource is forbidden” or no lists appear.	Use the site URL (not the list URL) and sign in with Microsoft account , not Anonymous. Clear old credentials in File > Options and settings > Data source settings .
Dates are wrong or show errors.	Change the type using Using Locale... and choose English (Malaysia) or the locale that matches your data.

The screenshot displays the Power Query Editor interface. The main area shows a table with the following data:

	NewColumnName	CustomerID	EmployeeID	OrderDate	Request
1	10248	VINET	5	7/4/1996, 12:00:00 AM +00:00	8/1/
2	10249	TOMSP	6	7/5/1996, 12:00:00 AM +00:00	8/16/
3	10250	HANAR	4	7/8/1996, 12:00:00 AM +00:00	8/5/
4	10251	VICTE	3	7/8/1996, 12:00:00 AM +00:00	8/5/
5	10252	SUPRD	4	7/9/1996, 12:00:00 AM +00:00	8/6/
6	10253	HANAR	3	7/10/1996, 12:00:00 AM +00:00	7/24/
7	10254	CHOPS	5	7/11/1996, 12:00:00 AM +00:00	8/8/
8	10255	RICSU	9	7/12/1996, 12:00:00 AM +00:00	8/9/
9	10256	WELLI	3	7/15/1996, 12:00:00 AM +00:00	8/12/
10	10257	HILAA	4	7/16/1996, 12:00:00 AM +00:00	8/13/
11	10258	ERNSH	1	7/17/1996, 12:00:00 AM +00:00	8/14/
12	10259	CFNTC	4	7/18/1996, 12:00:00 AM +00:00	8/15/

The 'Query settings' pane on the right shows the 'Applied steps' list with 'Renamed columns' highlighted in red.

HANDS-ON EXERCISE 5.2

Build the Request Monitoring report

Objective	Create a one-page report that shows request volume, status, priority and trends with interactive filtering.
Estimated duration	60 minutes
Prerequisites	Exercise 5.1 completed.

Steps

1. In **Report view**, apply a theme: **View > Themes**, choose a simple theme, or ask the trainer for the course theme file (olive and terracotta).
2. Create the five measures from section 5.3: right-click the **Requests** table > **New measure**, type each formula and press **Enter**. Format **Completion Rate** as a percentage.
3. Add a text box at the top: **PC MTBE Process Improvement Requests – Fictional Training Data**.

4. Add four **Card** visuals across the top for Total Requests, Open Requests, Completion Rate and Avg Days to Complete.
5. Add a **Clustered bar chart**: Y-axis = Department, X-axis = Total Requests. Title: **Requests by department**.
6. Add a **Donut chart**: Legend = Status, Values = Total Requests. Title: **Current status of requests**.
7. Add a **Stacked column chart**: X-axis = Request Type, Y-axis = Total Requests, Legend = Priority.
8. Add a **Line chart**: X-axis = Submission Month, Y-axis = Total Requests. Title: **Requests submitted per month**.
9. Add **Slicers** for Department and Priority on the left side.
10. Add a **Table** with Request ID, Request Title, Owner, Priority and Status. Filter the visual to exclude Completed and Rejected so it lists open items only.
11. Click a department bar and observe how the other visuals cross-filter. Save the file as **PIR Request Monitoring.pbix**.

Expected result

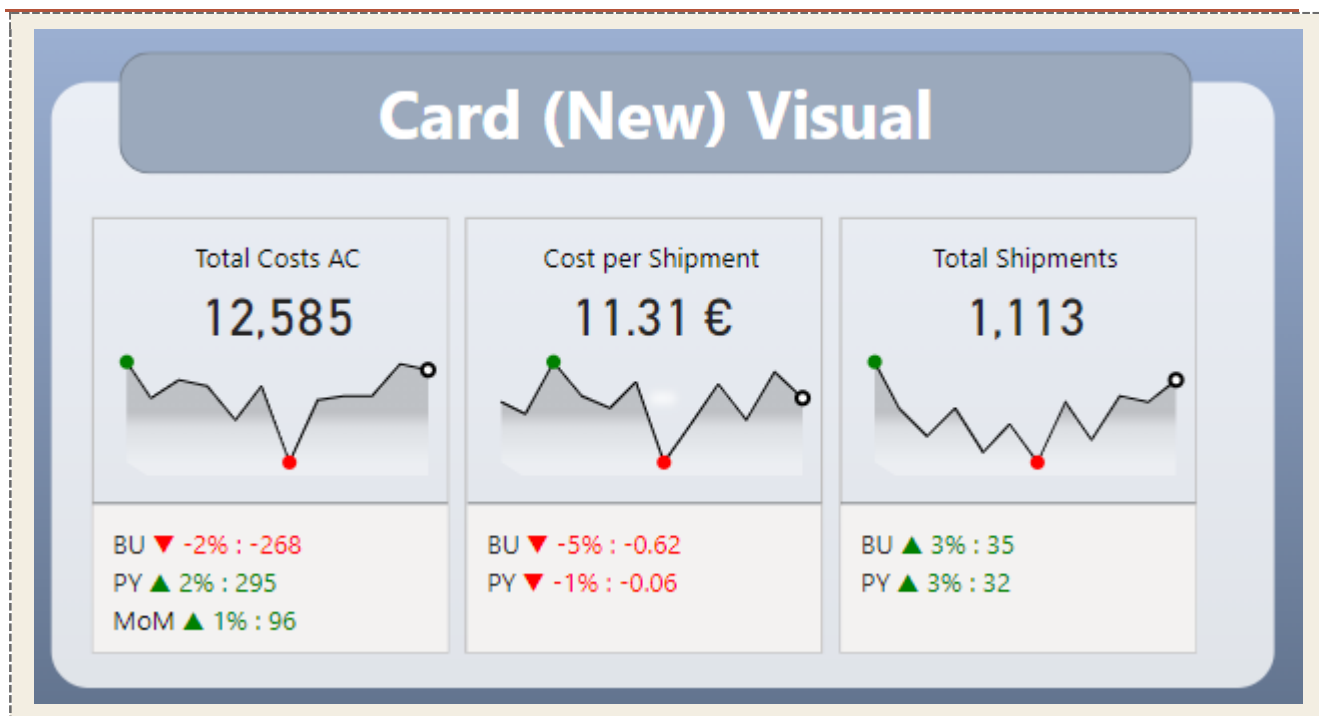
A clear, interactive single-page report with KPI cards, four charts, two slicers and an open-requests table, all based on fictional training data.

Checkpoint

- ☐ Total Requests matches the number of items in the SharePoint list.
- ☐ Selecting a department filters every other visual.
- ☐ Visual titles are written in plain language.

Common errors and how to resolve them

Common error	How to resolve it
The line chart shows one point or a year-only value.	Use Submission Month and, in the X-axis field drop-down, choose the column itself instead of Date Hierarchy .
Completion Rate shows 0.25 instead of 25%.	Select the measure, then on the Measure tools ribbon set the format to Percentage .



5.5 Publishing and sharing

When the report is ready, select **Home > Publish** in Power BI Desktop and choose a **workspace**. The report and its semantic model are uploaded to the Power BI Service. Because SharePoint Online is a cloud source, you can set **scheduled refresh** in the semantic model settings (after entering credentials) without installing a gateway.

- **My workspace** is personal. Use it for practice only — not for sharing with teams.
- **Shared workspaces** have roles: **Admin**, **Member**, **Contributor** and **Viewer**. Report consumers should normally get access through an App rather than a workspace role.

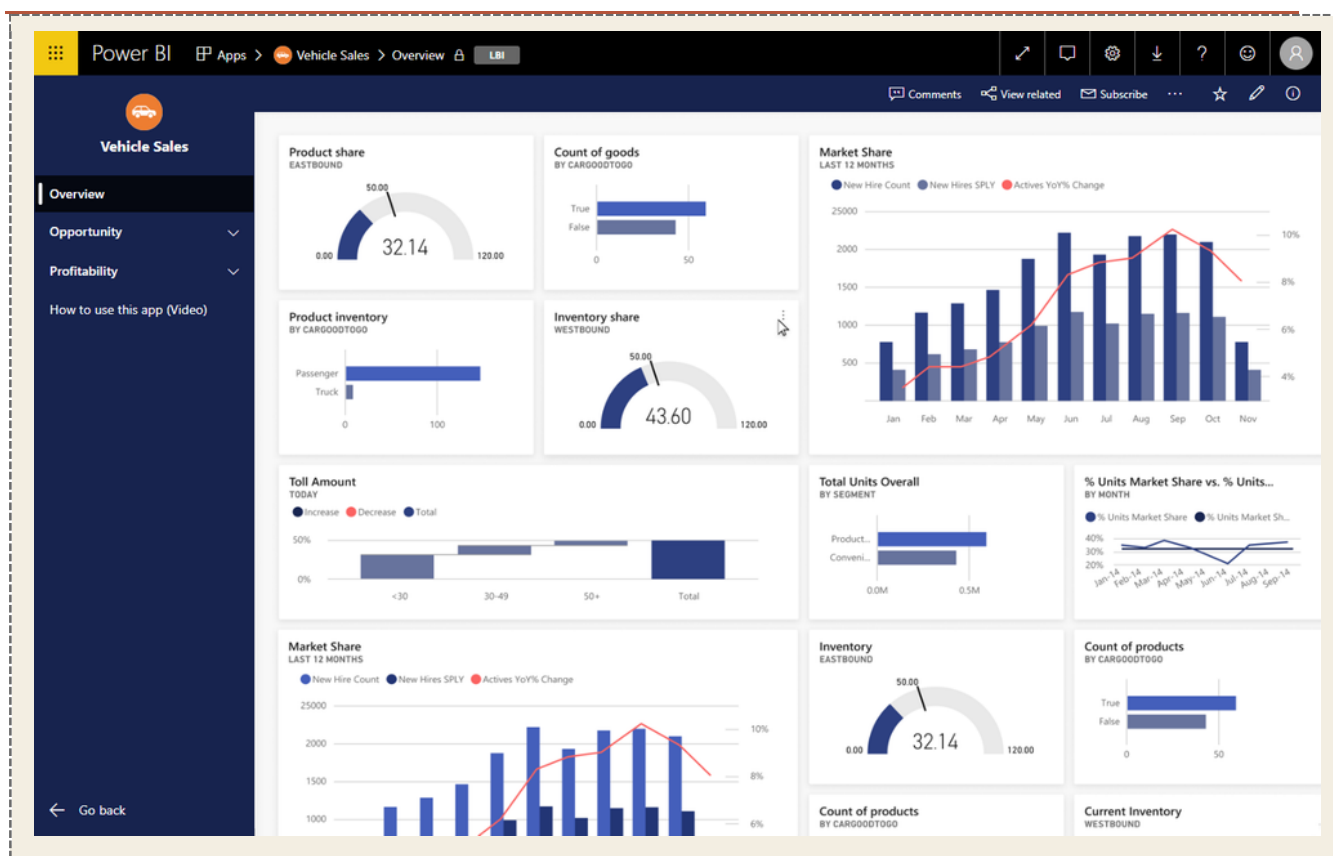
CHECK WITH YOUR ADMINISTRATOR

Publishing to a shared workspace and sharing content normally require a **Power BI Pro** or **Premium Per User (PPU)** licence for the publisher. Viewers also need Pro/PPU unless the workspace is on **Power BI Premium or Microsoft Fabric capacity (F64 or higher)**. Your organisation also controls who can create workspaces and apps. Do not assume you can publish or share — check with your administrator.

5.6 The Power BI App: one place for related reports

A **Power BI App** gathers related reports and dashboards from one workspace into a single, easy-to-navigate package. Instead of sending managers several links, you publish one app — for example, **PC MTBE Improvement Insights** — containing the Request Monitoring report, a Department Performance report and a management dashboard.

- **Consistent experience**: users see a clean navigation pane with sections and links, not the workspace behind it.
- **Controlled changes**: you can improve reports in the workspace and only update the app when ready.
- **Audiences**: one app can show different content to different groups (for example, managers versus HSE committee).
- **Remember**: a Power BI App distributes reports. It is **not** Power Apps, which builds data-entry applications.



HANDS-ON EXERCISE 5.3

Publish reports and package them into a Power BI App

Objective	Publish the report to a workspace and bundle two related reports into one Power BI App for a defined audience.
Estimated duration	25 minutes (hands-on if licensed; otherwise follow the trainer's demonstration)
Prerequisites	Exercise 5.2 completed; Power BI Pro/PPU licence and Member or Admin role in a training workspace. Check with your administrator.

Steps

1. In Power BI Desktop, select **Home > Publish** and choose the training workspace (for example, **PC MTBE Improvement - Training**).
2. Create a second report for the app: select **File > Save as PIR Department Performance.pbix**, replace the page with a matrix of Department by Status and a bar chart of Avg Days to Complete by Department, then **Publish** it to the same workspace.
3. In the Power BI Service (app.powerbi.com), open the workspace and select **Create app**.
4. On **Setup**, enter the name **PC MTBE Improvement Insights**, a short description, an app colour and optionally a logo.
5. On **Content**, select **Add content** and add both reports. Use **Add section** to create sections such as "Overview" and "Departments", and arrange the order.
6. On **Audience**, add the people or security group allowed to view the app (for the exercise, add the trainer or a colleague). Leave extra permissions such as build or reshare off unless required.
7. Select **Publish app**, then copy the app link and open it to check the navigation.

Expected result

A published Power BI App named PC MTBE Improvement Insights with two reports organised into sections and available to the chosen audience.

Checkpoint

- ☐ Both reports appear in the workspace.
- ☐ The app opens with your name, colour and sections.
- ☐ You can explain why managers receive the app link rather than workspace access.

Common errors and how to resolve them

Common error	How to resolve it
Create app is not visible.	You need Member or Admin role in the workspace, and app creation must be allowed by your tenant settings. Check with your administrator.
Audience members see "You need a Power BI Pro licence".	Viewers need Pro/PPU unless the workspace is on Premium or Fabric capacity. Check with your administrator.

MY NOTES

SECTION 6

Integrated exercise and review

Bring everything together and test the complete digital process from start to finish.

HANDS-ON EXERCISE 6.1

End-to-end test of the Process Improvement Request solution

Objective	Run one request through the full workflow and confirm that each platform responds correctly.
Estimated duration	40 minutes
Prerequisites	All exercises in Sections 2 to 5 completed; your flows are turned on.

Steps

- 1. Submit (Power Apps):** open your published app and submit a new request — Department: HSE; Type: Digital Form; Title: **Near-miss reporting form**; Priority: High; with a meaningful description. (Fictional training data.)
- 2. Store (SharePoint):** open the list and confirm the item exists with Status = Pending Approval and a Request ID.
- 3. Notify (Power Automate):** confirm you received the new-request email and the approval request.
- 4. Approve:** approve the request with the comment **Proceed – pilot at one area first**. Confirm the list shows Approved and the requester email arrived.
- 5. Update (Power Apps):** in the app, change the Status to **In Progress** and assign Owner = **Nurul Huda**.
- 6. Analyse (Power BI):** in Power BI Desktop select **Home > Refresh** (or refresh the published semantic model). Confirm the new request appears in the totals, the HSE bar and the open-requests table.
- 7. Distribute (Power BI App):** if you published an app, open it and confirm the updated figures after the semantic model refresh. If you updated report design, select **Update app** in the workspace.
- 8. Reflect:** write down one manual process in your own department that could follow the same pattern.

Expected result

The request moves from submission to approval and In Progress, with automatic ID, notifications and status updates, and it appears in the Power BI report after refresh.

Checkpoint

- ☐ Each of the seven platform steps worked without manual copying of data.
- ☐ You can explain which platform performed each step.
- ☐ You have identified one real process improvement idea.

Common errors and how to resolve them

Common error	How to resolve it
The new request is missing in Power BI.	Power BI shows data as at the last refresh. Refresh the report or semantic model, then check any report-level filters.

Common error	How to resolve it
No approval email arrived.	Open the flow's run history. If the flow is off or failed, correct the step shown in red and resubmit a test request.

Participant checklist

☐	I can explain the difference between a SharePoint site, a document library and a SharePoint List.
☐	I can view and restore a previous version of a document.
☐	I created the Process Improvement Requests list with appropriate column types and views.
☐	I generated a Canvas App from the list and customised the gallery and form.
☐	My app validates required information and shows clear success and error messages.
☐	I built a flow that assigns a Request ID and sends a notification.
☐	I built a basic approval that updates the status to Approved or Rejected.
☐	I built a scheduled reminder and tested it manually.
☐	I connected Power BI to the list, prepared the data and created measures.
☐	I built a report showing volume, status, priority and trends.
☐	I can explain the difference between a Power BI App and Power Apps.
☐	I know which steps require licences or permissions and whom to ask.

Review questions

Answer each question on your own first, then compare with the answers that follow.

Q1. Which platform in our case study stores the request records, and why is it a good choice for beginners?

Q2. What is the difference between a Power BI App and Power Apps?

Q3. Name the three screens generated automatically when you create a Canvas App from a SharePoint List.

Q4. Why did we use the trigger “When an item is created” instead of “When an item is created or modified”?

Q5. In the approval flow, what does the Condition check, and what happens in each branch?

Q6. Why should you prepare data in Power Query rather than by editing it manually?

Q7. What is a measure, and give one example from the course.

Q8. Give two situations in which you should check with your administrator.

Answers to review questions

A1. The **SharePoint List**. It is part of Microsoft 365, easy to create, supports column types and permissions, and connects to Power Apps, Power Automate and Power BI through standard connectors.

A2. A **Power BI App** packages and distributes Power BI reports and dashboards for viewing. **Power Apps** is used to build business applications that capture and update data.

A3. **BrowseScreen1** (gallery), **DetailScreen1** (display form) and **EditScreen1** (edit form).

A4. Because the flow updates the same item. With “created or modified”, each update would trigger the flow again, creating a loop.

A5. It checks whether **Outcome** equals **Approve**. If yes, Status becomes Approved and the requester is notified; if no, Status becomes Rejected and the comment is emailed to the requester.

A6. Power Query records each change as an **Applied Step**, so the same cleaning is repeated automatically every time the data refreshes.

A7. A reusable DAX calculation that responds to filters, e.g. **Total Requests = COUNTROWS(Requests)** or **Completion Rate**.

A8. Examples: publishing or sharing in Power BI (licences and capacity), using premium connectors, sharing apps, creating sites or lists, or when a connector is blocked by DLP policy.

APPENDIX A

Glossary of essential terms

Term	Meaning
Action	A step in a Power Automate flow that performs a task, such as sending an email.
Applied Steps	The recorded list of data preparation steps in Power Query.
Approval	A Power Automate feature that sends a request to one or more people to approve or reject.
Canvas App	A Power Apps application whose screens you design freely and connect to data.
Choice column	A SharePoint column that limits entries to a fixed list of options.
Condition	A decision in a flow that follows a Yes or No branch.
Connector	A prebuilt link between Power Platform and a service such as SharePoint or Outlook. Standard or premium.
DAX	Data Analysis Expressions — the formula language for Power BI measures and calculated columns.
Delegation	Passing filtering and sorting to the data source so an app can work with large lists.
Document library	A SharePoint container for files, with version history, metadata and permissions.
Dynamic content	Values from earlier steps in a flow that can be inserted into later steps.
Environment	A container in Power Platform for apps, flows and data, often separated into training, test and production.
Gallery	A Power Apps control that displays a list of records.
Measure	A DAX calculation that recalculates based on filters and selections.
Power BI App	A packaged bundle of Power BI reports and dashboards distributed to an audience.
Power Fx	The Excel-like formula language used in Power Apps.
Power Query	The tool used to connect to, clean and shape data in Power BI and Excel.
Run history	The record of each flow run, showing success or failure details.
Semantic model	The data, relationships and measures behind a Power BI report (previously called a dataset).
SharePoint List	A shared table of items with typed columns, views and permissions.
Trigger	The event that starts a flow, such as a new list item or a scheduled time.
Version history	The saved record of earlier versions of a file or list item, which can be viewed or restored.
Workspace	A collaborative area in the Power BI Service where content is created and managed.

APPENDIX B

Quick troubleshooting reference

Platform	Problem	Quick fix
SharePoint	Cannot create a list or share a file	Request Edit/Owner permission from the site owner; sharing may be restricted by policy
SharePoint	Pasted values are rejected in grid view	Match choice values exactly; check date format and column order
Power Apps	New column missing from the form	Refresh the data source, then add the field through Edit fields
Power Apps	Formula shows a red error	Check control names, brackets, commas and straight quotation marks
Power Apps	Users see an old version	Publish the app; users close and reopen it
Power Automate	Flow does not trigger	Confirm the flow is On, and the trigger site and list are correct
Power Automate	Condition goes to the wrong branch	Compare exact values (e.g. Approve) and check data types
Power Automate	Connection error or flow suspended	Fix the connection under Data > Connections; check DLP policy with your administrator
Power BI	Cannot connect to the list	Use the site URL and sign in with Microsoft account; clear old credentials
Power BI	Choice columns show "Record"	Expand the column and select Value in Power Query
Power BI	Cannot publish or create an app	Licence, workspace role or tenant setting is missing — check with your administrator

TRAINER TIP

When asking for help, share the **exact error message**, a screenshot, the name of the app or flow, and what you were trying to do. This makes it much faster for colleagues or IT to support you.

APPENDIX C

Action planning and notes

Use this page to plan how you will apply what you learned. Choose a small, realistic process first — early success builds confidence and support.

Planning question	My answer
Which manual process in my work takes the most time or causes the most errors?	
Where is the data kept today (email, Excel, paper)? Could it move to a SharePoint List?	
What would a simple app need to capture?	
Which notifications, approvals or reminders could be automated?	
What would my manager want to see in a Power BI report?	
Whom do I need to involve (process owner, IT, administrator)?	
My first step and target date	

MY NOTES

NOTE

Thank you for participating. For follow-up questions about this programme, contact the trainer at fakhrulsyahmi@mufasyahconsultant.com.